

Channel Position Estimation via Optical Flow

A CPU-only, Layered Approach to Robust Sensor Localization

Technical Report Summary

Pipe Inspection Pipeline

May 5, 2026

Introduction

Problem

For each frame of a channel inspection video, estimate the **absolute position** of the traveling sensor, the **turning-point frame** (where it reverses), and the **per-frame movement direction**.

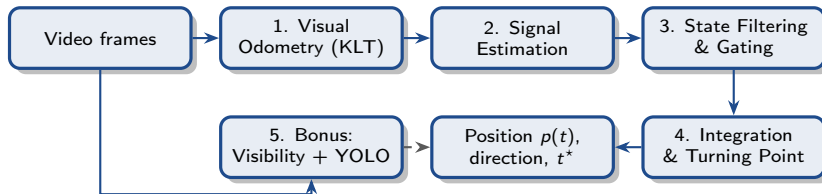
Design Targets

- ▶ CPU-only — no neural network, no GPU
- ▶ ~1000 frames in < 8 s (≈ 6 s typical, dominated by I/O)
- ▶ Robust to outdoor setup phase, reflections, motion blur
- ▶ Graceful μ s-fallback (linear triangle) if OpenCV is unavailable

Key Idea

Combine three *independent* cheap visual signals and gate them *before* integration — so that no single noisy signal drives the final position estimate.

The Pipeline



Why this ordering

- ▶ Cheap signals first → heavy work only on a downsampled subset.
- ▶ Gating *before* integration → outdoor frames cannot accumulate spurious distance.
- ▶ Turning point on *integrated* motion → uses full motion budget, not single-frame signals.

1. Visual Odometry Engine

Sparse Lucas–Kanade (KLT) at reduced resolution

- ▶ Grayscale + downscale to **25%** ($1080p \rightarrow \sim 480 \times 270$)
- ▶ Temporal stride **8**: only every 8th frame is a keyframe
- ▶ Shi–Tomasi corners: up to 240 pts, quality 0.008, min-dist 6 px
- ▶ 3-level pyramid, 25×25 window, ≤ 25 KLT iterations

Robustness decisions

- ▶ **Sparse, not dense**: avoids paying for textureless pipe regions
- ▶ **Permissive corner params**: tolerate texturally sparse pipe walls
- ▶ **Pyramid + 25 iter.:** handle motion blur and brightness drift
- ▶ **Linear interp. between keyframes**: valid because inspection motion is slow and continuous

Cost

$\sim 2\text{--}4$ ms per flow call. Downscale + stride $\rightarrow \approx 32\times$ fewer evaluations than dense full-resolution.

2. Signal Estimation — Motion Signals

Flow speed (motion magnitude)

- ▶ Exclude points within 5% of image center (radial flow ≈ 0 on the optical axis)
- ▶ MAD inlier filter: keep $|m_i - \text{med}| \leq 4 \text{ MAD} + 1.0$
- ▶ Speed = *median* of inlier magnitudes / frame gap

Why: the double-median (characterize then summarize) is resistant to static background features *and* reflection / motion-blur outliers.

Radial flow direction

- ▶ Signed projection of flow on the radial axis (away from center)
- ▶ Forward motion $\rightarrow$ expanding (+); reverse $\rightarrow$ contracting (-)
- ▶ Same MAD inlier set as speed

Why: a sign change is a direct, physically motivated proxy for the turning event.

2. Signal Estimation — Scene Signals

Pipe-likeness score (color heuristic)

- ▶ On 256×192 thumbnail: yellow-green pipe-color fraction, central dark-bore fraction, blue-sky fraction
- ▶ $\text{score} = 2.0 p_{\text{pipe}} + 0.8 p_{\text{dark}} - 1.0 p_{\text{blue}}$

Why: positive terms reinforce inside-the-pipe context; the sky penalty actively suppresses the outdoor setup phase at the start of each video.

Median-image NCC

- ▶ Median image from up to 600 sampled frames $\approx$ typical pipe interior
- ▶ Per-frame normalized cross-correlation, rescaled to $[0, 1]$

Why: a *motion-independent* scene-type indicator. Outdoor / setup frames differ from the median and score low; inside-pipe frames score high. Complements the color heuristic.

3. State Filtering and Gating

Three sequential gates — applied *before* integration

- ➊ **Pipe-interval mask:** smooth pipe-score (41-frame MA), threshold at $\max(P_{45}, 0.8)$, keep contiguous interior ± 16 frames
- ➋ **Startup / end masks (NCC):** release at first run of ≥ 100 frames whose NCC exceeds $\max(P_5, 0.7 \text{ med})$; mirrored at the end
- ➌ **Speed gate + noise floor:** 31-frame MA, re-mask, subtract $0.4 \cdot P_{10}$ of positive speeds

Why three gates and not one

- ▶ Each signal has a *different* failure mode — combining them removes outdoor frames the others miss
- ▶ **100-frame minimum run:** prevents a single similar-looking outdoor frame from prematurely opening the gate
- ▶ **Re-masking after smoothing:** stops moving averages from bleeding signal into already-gated regions
- ▶ **Noise floor:** prevents slow baseline drift from accumulating into spurious displacement

4. Integration and Turning Point Detection

Two-stage turning point

- 1 **Coarse:** frame at which cumulative motion first reaches 50% of total. Maximum-likelihood under symmetric round-trip; uses the *full* motion budget.
- 2 **Refined (still-flow):** pause segments where joint proxy $\max(\hat{s}, |\hat{d}|)$ stays below adaptive threshold $\theta \in [0.03, 0.35]$ for ≥ 5 frames. Score:

$$\text{score} = \text{still_strength} \cdot \text{length_bonus} \cdot (1 - 0.35 \text{ dist_penalty})$$

Accept iff $\text{score} > 0.08$, $\text{length} \geq \max(8, 0.6\% N)$, $\text{median} \leq 0.6 \theta$.

- 3 **Fallback:** radial-direction sign change in narrower window, scored by left/right-median contrast.

Dual-leg normalization (scale-free)

Outbound: cumulative motion stretched $0 \rightarrow L$ up to t^* . Inbound: stretched $L \rightarrow 0$ from t^* to end.

Why: removes dependence on unknown sensor-to-wall distance and lens focal length — the absolute pixel-scale of optical flow becomes irrelevant.

5a. Bonus — Rule-Based Visibility (BonusVideoAnalyzer)

Pipeline

- ▶ 10 features on 256×144 thumbnails: brightness, contrast, Laplacian sharpness, edge density, feature count, saturation, glare, dark ratio, color cast, Δframe
- ▶ Clarity score: $0.35 e + 0.30 f + 0.20 s + 0.15 c$
- ▶ 5 states: *clear*, *low_visibility*, *lens_covered*, *splash_water*, *under_water*
- ▶ Label-flip smoothing + min-duration filter (0.5 s transient / 1.0 s persistent)

Why rule-based is the right tool here

- ▶ Handles **non-spatial** states — blur, submersion, flat darkness — where any object detector would see no edges
- ▶ Stride 10 $\rightarrow$ ~ 300 samples on a 3000-frame video, < 3 s on CPU
- ▶ Output events are semantically meaningful: *lens_covered* $\rightarrow$ clean camera; *under_water* $\rightarrow$ flooded; *splash_water* $\rightarrow$ transient disturbance

5b. Bonus — Zero-Shot Defects (YoloBonusAnalyzer)

Pipeline

- ▶ YOLO-World, open-vocabulary prompts; no task-specific fine-tuning
- ▶ 10 defect classes (roots, gravel, sludge, grease, cracks, ...) + 4 visibility prompts
- ▶ Stride 25; same-class events within 2-stride frames merged into one event
- ▶ Setup-window suppression: 8% of frames (clamped 150–500) discarded

Complementarity with the rule-based analyzer

- ▶ YOLO handles **localized structural** defects the rule-based classifier cannot tell from normal texture
- ▶ Rule-based handles **non-spatial** states YOLO cannot detect
- ▶ Combined into a 4-level taxonomy: *clean* / *dirty* / *possibly_problematic* / *unusable*
- ▶ **Prompt design matters:** “thin roots growing into a sewer pipe” beats “roots” — reduces false positives from other thin elongated structures

Robustness comes from layering, not from any single model

- ▶ **Resource cost:** 25% downscale, stride-8, sparse flow, 256×192 thumbnails
- ▶ **Path accuracy:** three independent gated signals + dual-leg normalization (scale-free)
- ▶ **Turning-point accuracy:** integrated-motion coarse + still-flow refinement + sign-change fallback
- ▶ **Diagnostics:** rule-based visibility + zero-shot YOLO-World cover the full failure spectrum